
From RANS Prediction to Hydrofoil Optimization: A Benchmark of Neural Surrogates

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Abstract

1 We benchmark four neural surrogates for hydrofoil RANS prediction and
2 design: CNN-U-Net, a physics-regularized point network, a Fourier neural
3 operator (FNO), and DeepONet. The dataset contains 381 valid Open-
4 FOAM cases. The 64-case validation set consists of two NACA families
5 that are absent from training. The models predict flow fields, force coeffi-
6 cients, and pressure-based cavitation risk. The physics-regularized network
7 gives the best pressure result ($R^2 = 0.966$) and cavitation-risk F1 score
8 (0.921). DeepONet gives the best lift and drag results ($R^2 = 0.996$ and
9 0.988). Inference is 917–1801 times faster than OpenFOAM. We then place
10 each model in the same shape optimizer and rerun its best design in Open-
11 FOAM. The validated lift-to-drag improvements range from 2.3% to 79.7%.
12 The benchmark connects predictive accuracy and computational cost to the
13 quality of the design produced downstream.

14 1 Introduction

15 Computational fluid dynamics (CFD) supports hydrofoil analysis and design, but repeated
16 Reynolds-averaged Navier–Stokes (RANS) solves make parameter sweeps and optimization
17 expensive. Neural surrogates can reduce this repeated cost [Brunton et al., 2020]. A useful
18 design surrogate must do more than match an average flow field. It must preserve pressure
19 and forces, identify low-pressure regions, and guide an optimizer toward a design that still
20 performs well when it is returned to CFD.

21 CNN-U-Nets learn local multiscale structure Ronneberger et al. [2015]. Fourier neural op-
22 erators learn global spectral mappings Li et al. [2021], Kovachki et al. [2023]. DeepONet
23 combines parameter and coordinate representations Lu et al. [2021], while physics-informed
24 models add differential-equation residuals Raissi et al. [2019]. These architectures make
25 different assumptions about the flow map. Comparing them on one dataset reveals where
26 each assumption helps.

27 Earlier airfoil surrogates used convolutional or hybrid networks to predict steady flow fields
28 from geometry and operating conditions Guo et al. [2016], Bhatnagar et al. [2019], Sekar
29 et al. [2019], Thuerey et al. [2020]. Physics-informed neural operators combine PDE residuals
30 with operator learning Li et al. [2024]. We instead hold the data and optimizer fixed and
31 compare how four model families perform from flow prediction through design selection.

32 We use 381 valid OpenFOAM simulations covering twelve NACA families, eight angles of
33 attack, and four Reynolds numbers. All models use the same targets and unseen-geometry
34 validation set. We compare flow and force accuracy, cavitation-risk classification, runtime,
35 and continuous shape optimization. Each model-selected design is then rerun in OpenFOAM.

36 This final test asks a simple question: does the surrogate lead the optimizer to a better
37 hydrofoil?

38 Our contributions are:

- 39 • a four-model benchmark on an unseen-geometry validation set, covering flow fields,
40 force coefficients, runtime, and cavitation-risk screening;
- 41 • a common hydrodynamic shape-optimization experiment; and
- 42 • an OpenFOAM rerun of every model’s selected design.

43 We intentionally make a limited cavitation claim. The present labels measure pressure-
44 threshold inception risk derived from single-phase RANS pressure. They do not model
45 vapor fraction, cavity length, collapse, or shedding, which require multiphase simulations or
46 experiments.

47 2 Data and CFD Reference

48 2.1 Design space

49 The dataset contains NACA 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418,
50 and 4421 profiles generated from the standard four-digit parameterization Abbott and von
51 Doenhoff [1959]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$
52 and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. The chord is 1 m, water density
53 is 997 kg/m³, and kinematic viscosity is 10⁻⁶ m²/s, so the freestream speed is set by
54 $U_\infty = Re\nu/c$. The far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

55 Each case is solved with OpenFOAM `simpleFoam` Weller et al. [1998], using steady in-
56 compressible RANS and the $k-\omega$ SST closure Menter [1994]. Angled freestream velocity is
57 imposed at the inlet and far boundary. The solver runs for 300 iterations, and OpenFOAM’s
58 `forceCoeffs` provides total lift and drag references. Cell-centered fields are interpolated
59 to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs
60 include coordinates, operating parameters, three NACA geometry descriptors, a fluid mask,
61 and signed distance to the profile.

62 An audit found no pressure fields duplicated across angle of attack and found force files for
63 all 384 cases. Three cases (NACA 0006 at 10°, NACA 0008 at -4°, and NACA 4412 at
64 0°) produced $|C_L|$ or $|C_D| > 5$ and were excluded before splitting. This leaves 381 cases.
65 The deterministic geometry split holds out all 64 cases from NACA 0015 and NACA 2412;
66 the remaining 317 cases train the models. Holding out complete families prevents identical
67 geometry from appearing at different operating conditions in both partitions.

68 2.2 Targets and cavitation-risk definition

69 The common output vector contains U_x , U_y , absolute pressure p , eddy viscosity, k , ω ,
70 C_p , cavitation margin, C_L , and C_D . Grid fields use masked normalized mean-squared
71 error, while the case-level force targets use normalized mean-squared error. OpenFOAM’s
72 kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \quad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}. \quad (1)$$

73 For an ambient-pressure value p_a , the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y)\frac{1}{2}\rho U_\infty^2 - p_v. \quad (2)$$

74 A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in$
75 $\{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test
76 from the predicted C_p field without representing multiphase physics.

77 3 Surrogates and Optimization

78 3.1 Model families

79 The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization,
80 GELU activations, and base width 16. The FNO has width 32, four operator blocks, and
81 12 retained Fourier modes per direction Ronneberger et al. [2015], Li et al. [2021]:

$$h_{l+1} = \sigma(W_l h_l + \mathcal{F}^{-1}(R_l \mathcal{F}(h_l))). \quad (3)$$

82 The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions.
83 Its output has the form

$$\hat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x}) / \sqrt{64} + c_k. \quad (4)$$

84 following Lu et al. [2021]. The physics-regularized point network has three tanh layers of
85 width 64 and uses 1,024 sampled points per case. Its supervised loss is augmented by 10^{-4}
86 times the continuity and first-order momentum residual loss Raissi et al. [2019]. This is a
87 lightweight physics regularizer, not a data-free PINN or complete RANS residual.

88 All models use AdamW with learning rate 2×10^{-3} , weight decay 10^{-4} , at most 120 epochs,
89 patience 20, gradient clipping, and seed 7. Normalization uses training cases only. The best
90 checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters
91 (Table 1).

92 Although surface-pressure integration was implemented as a diagnostic, the coarse common
93 grid did not reproduce absolute OpenFOAM forces reliably because it omits viscous traction
94 and under-resolves the wall. Therefore, optimization uses dedicated C_L and C_D output
95 heads trained on OpenFOAM’s total force coefficients. This choice avoids presenting an
96 invalid pressure-only force calculator as ground truth, but it also means force prediction is
97 directly supervised.

98 3.2 Common optimization protocol

99 At $Re = 5 \times 10^5$, every model optimizes the NACA variables $\theta = (m, p, t, \alpha)$ from NACA
100 0012, 2412, and 4415 starts. Bounds are $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and
101 $\alpha \in [-4, 10]^\circ$. Nelder–Mead Nelder and Mead [1965] runs for at most 40 iterations with
102 identical tolerances. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}} - 10P_M - 5f_M, \quad (5)$$

103 where P_M penalizes a predicted minimum cavitation margin below 5000 Pa and f_M is
104 the fraction of cells below that margin. At the nominal ambient pressure, all selected
105 designs remain far above the threshold, so the cavitation penalty is inactive; the separate
106 ambient-pressure sweep evaluates inception classification under reduced pressure. For each
107 architecture, we select the highest-scoring design across starts and create a new mesh and
108 OpenFOAM case without using surrogate predictions to initialize the solver.

109 4 Results

110 4.1 Validation accuracy and computational cost

111 Table 1 reports results on the 64-case validation set. The physics-regularized network is
112 best for pressure and C_p , with pressure RMSE 6.78 Pa and $R^2 = 0.966$. DeepONet gives
113 the best U_x , U_y , C_L , and C_D coefficients. The grid operators are weaker in this low-data
114 setting. FNO retains useful force trends, while CNN-U-Net predicts C_p and forces better
115 than absolute pressure or streamwise velocity.

116 Measured training times are 868, 104, 547, and 54 s for CNN-U-Net, the physics-regularized
117 network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is
118 12.4 s per case, while pressure-field inference requires 0.0069–0.0135 s, yielding 917–1801×

Table 1: Validation performance, model size, and cost. Force RMSE is computed case-wise against OpenFOAM total coefficients.

Model	p RMSE	p R^2	C_L RMSE	C_D RMSE	Params.	Speedup
CNN-U-Net	28.52	0.392	0.0991	0.00453	485,178	1,148×
Physics-reg. MLP	6.78	0.966	0.0245	0.00204	9,930	1,801×
FNO	22.73	0.614	0.1383	0.00957	595,914	1,028×
DeepONet	8.55	0.945	0.0268	0.00184	100,746	917×

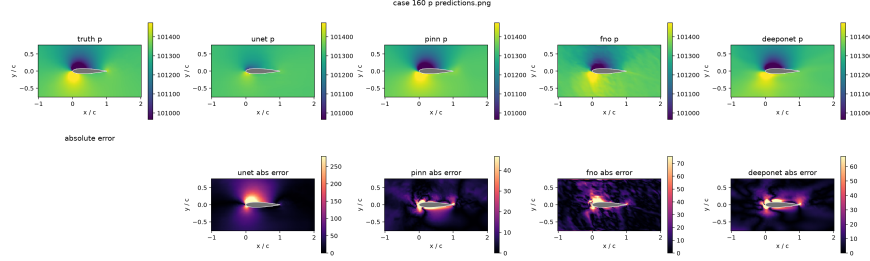


Figure 1: Absolute-pressure prediction for a representative validation case from the NACA 0015 case (case 160), followed by per-model absolute error. All prediction panels share the truth color scale; error panels use individually labeled scales.

speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on CPU and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

Figure 1 shows a representative validation case from the unseen NACA 0015 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. The distinct error color scales in the lower row are retained to expose each model’s error structure rather than visually equalizing models with very different ranges.

4.2 Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so accuracy is not useful here. The physics-regularized network achieves precision 0.927, recall 0.916, and F1 0.921. DeepONet follows at F1 0.913. CNN-U-Net has the highest recall (0.958) but lower precision (0.796), while FNO obtains F1 0.771. Twenty-four validation case/ambient-pressure combinations contain at-risk cells. This measures pressure-based inception screening, not cavity dynamics. Multiphase labels are required to predict vapor fraction, cavity length, or shedding Brennen [1995].

4.3 CFD-revalidated optimization

Table 2 compares each model’s predicted winner with a fresh OpenFOAM solve. All four winners improve upon their associated starting foil in CFD, but by very different amounts. The physics-regularized network selects a thin, strongly cambered profile whose validated $L/D = 45.96$ is 79.7% above the NACA 4415 baseline. DeepONet obtains a 57.8% improvement and CNN-U-Net obtains 48.5%. FNO predicts the lowest-performing winner; revalidation gives only a 2.3% gain over its baseline.

The optimization test sharpens the field benchmark in two ways. First, DeepONet gives the smallest winner-level L/D discrepancy, consistent with its force accuracy, but the physics-regularized model finds the best validated design. Second, the FNO winner improves after CFD revalidation yet remains close to its baseline, showing that a conservative prediction error does not guarantee strong search guidance. Screening of five randomly perturbed

Table 2: Surrogate-selected winners and independent OpenFOAM revalidation. Improvement is relative to the corresponding start at the optimization condition.

Model	Start	Pred. L/D	CFD L/D	Error	CFD gain
CNN-U-Net	2412	30.17	38.45	−21.5%	48.5%
Physics-reg. MLP	4415	43.62	45.96	−5.1%	79.7%
FNO	4415	21.11	26.15	−19.3%	2.3%
DeepONet	2412	39.99	40.86	−2.1%	57.8%

profiles and an oval was also performed, but these geometries lie outside the NACA training manifold and are reported only as stress tests, not validated free-form design evidence.

5 Scope and Next Steps

The pointwise models lead in this low-data, fixed-grid experiment. This is not a universal ranking. It is a controlled comparison in which every model sees the same data, targets, split, and optimizer. Larger datasets or different loss balancing may favor the grid operators.

Pressure, force, and optimization metrics answer different questions. DeepONet is the best force model, but the physics-regularized network returns the best CFD-validated design. This difference is the main reason to include optimization in the benchmark instead of stopping at validation error.

The current study covers two-dimensional, steady, single-phase RANS and NACA four-digit profiles. Three divergent cases are removed by a fixed quality rule, and a symmetry audit finds a small lift bias. Force coefficients are directly supervised because the common grid cannot resolve surface forces accurately. One winner per model is rerun. Cavitation results cover inception risk, not vapor transport or shedding. Repeated seeds, mesh sensitivity, multiphase labels, and more CFD checks around each optimum are the clearest next steps.

6 Conclusion

Across four neural surrogates, the physics-regularized point network gives the strongest joint pressure, inception-risk, speed, and validated-design result, while DeepONet predicts forces most accurately. The $917\text{--}1801\times$ speedups make search practical, and the OpenFOAM reruns show how prediction differences carry into design. The optimization experiment makes the comparison more useful than a field-only ranking.

Artifact Availability

Code, configurations, split definitions, trained checkpoints, per-case metrics, optimization traces, and OpenFOAM validation cases are organized in an auditable repository. An anonymized archival link will be added before submission.

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